

Case Study 1:- Unauthorized network scanning using Nmap:-

Scenario:- A final year engineering student scans a software company's network's using Nmap without permission to identify open ports and running services.

Ethical Analysis:-

- The student intended to learn cybersecurity.
- However, scanning without authorization is unethical.
- Ethical hacking requires written permission from the system owner.

Legal Analysis (IT ACT 2008)

Applicable Sections:-

- Section 43 - Unauthorized access or damage to computer system.
- Section 66 - Computer related offences done when dishonestly (or) fraudulently.

Possible legal consequences:-

- Civil compensation
- Criminal Investigation
- Fine (or) imprisonment depending on the offences.

Correct procedure:-

- obtain return permission
- Define testing scope
- sign authorization documents
- perform testing only within approved limits.

Conclusion:-

Scanning a network without permission is both the unethical and may be illegal, even if not damage.

Case Study 2:- public disclosure of SQL injection vulnerability.

Scenario:-

- A researcher discovers a SQL injection vulnerability but immediately post it on social media before of information the company.

Ethical Analysis:-

- public disclosure before a fix is irresponsible
- It increases the risk of cyberattacks
- Researchers should protect users.

Responsible Vulnerability Disclosure:-

- Notify the organization privately
- gives sufficient time to fix the issue
- publish details only after the patch is released.

Risk of Premature Disclosure:-

- Customer data theft.
- Financial losses
- Reputation damage
- Legal consequences.
- Increased cybercrime.

Legal Perspective

⇒ possible IT Act Scenarios:-

- Section 43
- Section 66

Recommended Steps:-

- Contact the organization's security team
- Share technical evidence privately.
- Wait for remediation
- publish research only after the vulnerability is fixed

Conclusion:-

- Responsible disclosure protects both organization and users.

Case study 3 :- phishing attack on financial organization scenario:-

- An employee receives a fake email pretending to be IT Support and enters login credentials

Type of cybeattack:-

Phishing Attack

- Social Engineering
- Credential theft.

Legal Implications (IT Act)

Applicable ~~sections~~:-

- Section 43 → unauthorized access
- section 66 → computer related offences
- section 66C → Identity theft.
- Section 66D → cheating by personation using computer resources.

Responsibilities:-

Organization:-

- Employee awareness training
- Email Security
- Multi factor Authentication
- Regular security audits.

Employee:-

- Verify suspicious email's
- Never share passwords
- Report phishing immediately
- Follow cybersecurity policies.

Organizational:-

- Cyber security awareness training
- phishing simulation
- password policy
- Incident response plan.

Conclusion:-

Both employee's and organizations share responsibility for preventing phishing attacks.

Case Study 4:- Student Exploits Banking Vulnerability

Scenario:-

- A cybersecurity student finds a banking vulnerability and retrieves a small amount of customer data before operating it.

Ethical Analysis:-

- Good intentions does not justify unauthorized exploration
- Accessing customer data violates privacy.
- Ethical hackers must avoid accessing sensitive information.

Ethical hacking	Vs	Unauthorized Access
permission obtained		No permission
legal		illegal
Improve Security		violated security
Controlled testing		unauthorized exploration

Applicable IT Act Sections! -

→ Section 43

→ Section 66

→ Section 72.

proper Reporting process! -

→ Do not exploit the vulnerability

→ collect minimal evidence

→ Inform the bank privately

→ Request acknowledgement.

→ wait for the organization to fix issue.

→ publish only after permission (or) remediation.

Conclusion! -

Even with good intentions, unauthorized exploration of vulnerability can lead the legal action.